

SatAI-ForestWatch: Scalable Deep Learning for Real-Time Forest Monitoring in India

ABSTRACT

Deforestation remains a critical environmental challenge in India, with severe impacts on biodiversity, carbon sequestration, and sustainable development. Timely and accurate detection of forest cover loss is therefore essential for effective conservation planning and policy interventions. This study presents SatAI- Forest Watch, a scalable deep learning framework for near real-time deforestation detection using multi- temporal Sentinel-2 imagery. Sentinel-2 offers high spatial resolution (10 m) and rich spectral information across 13 bands, enabling robust monitoring of forest cover dynamics. Multi-seasonal composites were generated through Google Earth Engine to capture temporal variations in vegetation, while deep learning architectures were employed for semantic segmentation and change detection across diverse forest regions in India. Vegetation indices such as NDVI and NBR were integrated with multi-temporal stacks to enhance sensitivity to subtle disturbances.

Batch Number: Team_C_18C

Guide Name: Dr. K. Kanaka Vardhini

Ms. Sakshi Tapadia-24WH8A0501

Ms. Gopagoni Chaitanya-22WH1A05F5

Ms. Pentyala Niharika-22WH1A05F4